Spring potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate stored spring energy and distinguish it from force.

SUCCESS CRITERIA - I can...

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “compression” and “spring constant” to explain your answer. Sentence starters are optional.

compression: shortening or squeezing relative to an unstressed length

spring constant: how much an ideal spring’s force changes per stretch

potential energy: energy associated with the configuration of interacting objects

01 Warm-up: bring back an earlier idea

A pendulum loses **15 J** gravitational energy while descending. Neglect losses. **Find its kinetic-energy gain**.

Word help: pendulum = an object that swings about a fixed support

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

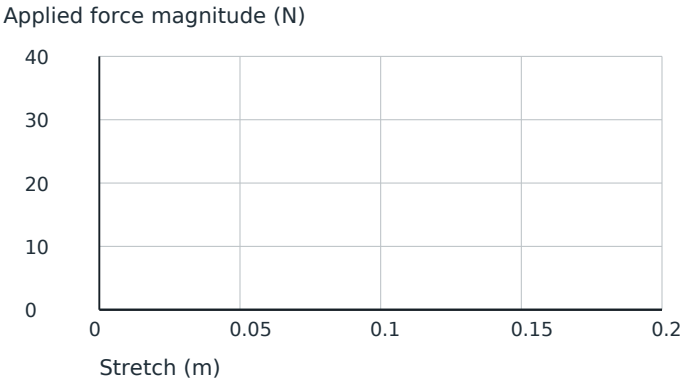
02 Worked example: explain the prediction

Reference: $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

Worked reasoning: A spring with k=**200 N/m** stretched **0.10 m** stores $\frac{1}{2}kx^2$=**1 J**. Doubling stretch to **0.20 m** stores **4 J**, not **2 J**.

Your turn: Draw force-versus-stretch triangles and **compare their areas**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The same spring is displaced x and **2** x. **Compare stored energies** and force magnitudes.

Sentence starter: Both ... . They differ in ... because ... .



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Apply the physics

1. Use k=**100 N/m** and x=**0.20 m**.
2. **Find x²**, then U=½×**100**×x².

Sentence starter: I found ... [unit]. This value means ... in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is energy positive for both positive and negative x even though spring force reverses?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 N** friction force opposes a **6 m** displacement. **Find work**, including sign and units.
Label force and displacement, including their relative direction.

Word help: work = energy transferred by a force along a displacement

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

07 Exit check

A spring stores **3 J** at a certain compression. **How much** does it store at double compression?

Sentence starter: I notice … . This supports … because … .



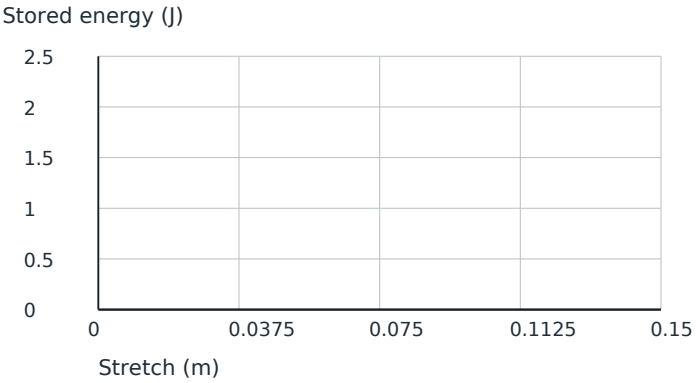
Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Find the energy** for stretches **0.05**, **0.10**, **0.15 m** when k=**200 N/m**.
2. **Sketch the energy trend**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3